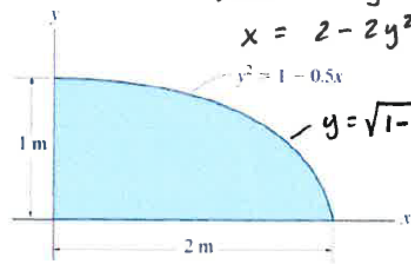
Practice POI (Ixy) by Integration Problem from an Old Exam 4

Easier to use calculator, the actual integration is longer….

$.5x = 1 - y^2$

$x = 2 - 2y^2$



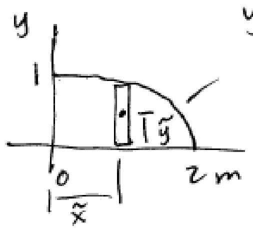
$y = \sqrt{1-.5x}$

3.2 Determine the product of inertia of the shaded area with respect to the x and y axes.

$$I_{xy} = \int_0^2 \int_0^{\sqrt{1-.5x}} xy \, dy \, dx$$

TI89: $\int(\int(x*y, y, 0, \sqrt{1-.5*x}), x, 0, 2)$

$= \boxed{.333 \; m^4 = I_{xy}}$



$y^2 = 1 - .5x$

$y = \sqrt{1-.5x}$

$x = 2 - 2y^2$

$$\boxed{I_{xy} = \int_A \tilde{x}\,\tilde{y}\, dA}$$

$\tilde{x} = x$

$\tilde{y} = \frac{1}{2} y$

$= \frac{1}{2}(1-.5x)^{.5}$

$dA = \; dx \sim y = (1-.5x)^{.5}$

$$\int_0^2 x \cdot \frac{(1-.5x)^{1/2}}{2} (1-.5x)^{1/2} \, dx$$

$$= .5 \int_0^2 x(1-.5x)\, dx$$

$$= .5 \int_0^2 x - .5x^2 \, dx$$

$$= .5 \left[\frac{1}{2}x^2 - \frac{.5}{3}x^3\right]_0^2$$

$$= .5 \left[2 - \frac{.5}{3}(8)\right]$$

$$= .5 \left[2 - 4/3\right]$$

$$= .5 \,(2/3)$$

$$= \boxed{1/3 \; m^4 = I_{xy}}$$